

The Blue Lotus Stress-Testing Engine: Design, Methodology, and Out-of-Sample Verification

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Abstract

We present the Blue Lotus Stress-Testing Engine, a system that turns a single return series into a forward-looking distribution of drawdown, tail-loss, and recovery outcomes. The engine couples a deterministic three-state volatility regime model, Bayesian shrinkage of regime parameters, Generalized Pareto tail estimation via Peaks-over-Threshold Extreme Value Theory, regime-switching Monte Carlo path generation with a drawdown-conditional blend and a hard loss floor, optional Esscher importance resampling, and fully bootstrapped confidence intervals on every reported metric. A Model Fragility Index, defined as the mean Wasserstein-1 distance between baseline and parameter-perturbed drawdown distributions, gives a single auditable measure of how much the risk estimates depend on uncertain inputs.

Beyond describing the method we subject it to a strict out-of-sample test. For nine named market crises and ten liquid asset proxies we re-fit the engine using only the history available before each crisis and compare its predicted drawdown distribution against the realized outcome, yielding 89 crisis cells; an identical protocol on 213 calm windows provides a calibration control. On calm data the engine's 5% and 1% tail bounds are breached at 2.8% and 0.9% rates respectively, neither rejectable by a Kupiec test ($p = 0.11$, $p = 0.93$); under the nine crises those rates rise to 56% and 31%, degrading monotonically with crisis severity and breaking completely only at the two fastest systemic shocks on record. The engine therefore reproduces ordinary risk faithfully and reports, rather than conceals, the residual tail gap that history alone cannot close.

Keywords: stress testing, Monte Carlo, regime switching, Extreme Value Theory, Expected Shortfall, Esscher transform, Wasserstein distance, out-of-sample backtest, Kupiec test.

JEL: C15, C53, C58, G17, G32.

Disclaimer: the engine produces probability distributions, not point predictions or investment advice. All outputs are conditional on the modelling assumptions described herein.

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1. Introduction

1.1 What the engine does, in plain terms

Give the engine a history of daily returns — a market index, a fund, or the profit-and-loss of a trading strategy — and it answers a single question: *if the world keeps behaving the way this series suggests it can, how bad could the next stretch get?* It does not forecast a price or a direction. Instead it simulates tens of thousands of plausible forward paths and reports the *distribution* of three things an allocator actually cares about: how deep a drawdown to expect, how heavy the losses are in the worst few percent of outcomes, and how long recovery tends to take. Every headline number arrives with a confidence interval, and the whole run can be exported as a clean JSON object or a formatted PDF.

1.2 Why ordinary Monte Carlo is not enough

The textbook approach draws returns from one fixed distribution and simulates forward. It is clean and it is wrong in two ways that matter for risk work.

First, returns are not drawn from a single stationary law. Volatility clusters, losses beget losses, and the probability of a large drop conditional on a recent run of drops is far higher than its unconditional value. A model blind to this structure understates sustained crisis drawdowns while overstating isolated single-day moves.

Second, tail data is scarce. A fifteen-year equity history holds roughly four thousand daily observations, of which only a few hundred belong to genuine stress. Any tail model fit to those few hundred points carries large estimation error, and plain simulation propagates that error silently, with no measure of how much to trust the answer.

1.3 Contributions

The Blue Lotus engine addresses both problems and adds an explicit honesty mechanism. Concretely, this paper makes four contributions:

1. A fully specified, auditable risk engine (Sections 2–4) that structures simulation around a deterministic volatility-regime model, estimates tails with Extreme Value Theory, stabilises parameters with Bayesian shrinkage, and reports bootstrap confidence intervals on every metric.
2. The Model Fragility Index (Section 4.8), a Wasserstein-based, scale-invariant measure of how sensitive the output distribution is to parameter uncertainty.
3. A reproducible single-asset characterisation (Section 6) and, centrally, a strict out-of-sample event-study backtest across nine crises and ten assets, with a 213-cell calm-period calibration control (Section 7).

4. An honest accounting (Sections 8–9) of where the engine is well calibrated and where, by construction, no history-fit model can reach.

2. The Engine at a Glance

The engine is a pipeline of seven stages, each a self-contained module with a typed input and output so that any stage can be inspected, replaced, or unit tested in isolation.

1. **Input processing.** Clean the return series, winsorise extremes, and record raw moments for reporting.
2. **Regime detection.** Label every day calm, volatile, or crisis from rolling realised volatility and a negative-trend gate.
3. **Bayesian shrinkage.** Pull the noisy per-regime means and variances toward pooled estimates in proportion to how little data supports them.
4. **Tail estimation.** Fit a Generalized Pareto Distribution to the loss tail via Peaks-over-Threshold and derive analytical VaR and ES targets.
5. **Monte Carlo generation.** Simulate regime-switching paths, blended with drawdown-conditional dynamics and clamped by a loss floor.
6. **Measure change and metrics.** Optionally tilt the path measure to hit mean and ES targets (Esscher), then compute the drawdown, ES, and recovery distributions with bootstrap confidence intervals.
7. **Fragility analysis.** Re-run the generator under parameter perturbations and summarise the output’s sensitivity as a single index.

The result is serialised to JSON and rendered to a PDF tear-sheet; the same core is exposed through a REST API and two front-ends (Section 5). Everything downstream of stage 1 is deterministic given a fixed random seed, so runs are exactly reproducible.

3. Design Rationale

Each modelling choice trades sophistication against auditability; the engine consistently favours methods whose outputs a risk officer can explain.

Deterministic regimes over a hidden Markov model. An earlier version used a three-state Gaussian HMM fit by Baum–Welch. It was theoretically attractive but produced crisis fractions that varied wildly across assets and needed manual reseeding to avoid degenerate states. The current detector assigns regimes by explicit rules on rolling volatility and trend (Section 4.2). Every label can be justified in one sentence, the crisis fraction is calibrated by construction through its percentile thresholds, and the model has no latent state to misconverge.

Extreme Value Theory for the tail. Parametric bulk distributions such as the Student- t constrain the extreme tail through the centre of the data. Peaks-over-Threshold EVT models only the exceedances, which is the asymptotically correct approach for extreme quantiles and the industry standard in quantitative risk management.

Bayesian shrinkage for stability. The crisis regime may hold only a few hundred days; its raw mean and variance are too noisy to drive a simulation. Conjugate shrinkage pulls each regime’s parameters toward pooled estimates by an amount that falls as the regime’s sample grows, trading a little bias for a large reduction in variance.

Esscher tilt as an optional corrector. When the simulated mean or ES drifts from the EVT and regime targets, an Esscher exponential tilt reweights paths to restore them. Crucially the tilt is optional and self-disabling: if its solver does not converge it falls back to identity weights, leaving the historical measure untouched rather than imposing an unstable correction.

Bootstrap intervals everywhere. A risk number without an interval invites false precision. The engine resamples paths to attach a 90% confidence band to every scalar it reports, so the user sees estimation uncertainty directly.

A fragility index for model risk. Even a well-calibrated model is only as trustworthy as it is stable. The Model Fragility Index quantifies how far the drawdown distribution moves under deliberate parameter perturbations, turning model risk into a reported number rather than an afterthought.

4. Methodology

4.1 Data preprocessing

Let $\{r_t\}_{t=1}^T$ be a daily return series. Returns are winsorised at the 1st and 99th percentiles before any estimation, clipping extreme observations to those boundaries to prevent a handful of points from dominating moment and tail estimates. Raw mean, standard deviation, skewness, and excess kurtosis are recorded *before* winsorising for reporting. Annualised volatility is $\hat{\sigma}_{\text{ann}} = \hat{\sigma}_{\text{daily}}\sqrt{252}$.

4.2 Regime detection

Market conditions are classified into three states: calm (0), volatile (1), and crisis (2). For each $t > w$ the engine computes the w -day annualised realised volatility and the w -day rolling return,

$$\hat{\sigma}_t = \sqrt{252} \widehat{\text{std}}(r_{t-w}, \dots, r_{t-1}), \quad \tilde{r}_t = \sum_{j=t-w}^{t-1} r_j, \quad (1)$$

with $w = 20$ by default. Let q_{70}, q_{90} be the 70th and 90th percentiles of $\{\hat{\sigma}_t\}$. The raw label is 0 below q_{70} , 1 in $[q_{70}, q_{90})$, and 2 at or above q_{90} .

Two filters refine the crisis state. A *negative-trend gate* downgrades any crisis candidate with $\tilde{r}_t > 0$ to volatile, so high-volatility recovery rallies are not mislabelled as crises. A *persistence filter* downgrades any run of crisis days shorter than $d_{\min} = 5$ consecutive days, so isolated spikes do not count as regimes. The first w observations are set to calm.

From the final labels $\{s_t\}$ the transition matrix is estimated with Laplace (add-one) smoothing,

$$P_{ij} = \frac{1 + \sum_{t < T} \mathbf{1}[s_t = i, s_{t+1} = j]}{3 + \sum_{t < T} \mathbf{1}[s_t = i]}, \quad (2)$$

per-regime means and standard deviations are sample moments over each state, and the long-run distribution $\boldsymbol{\pi}$ is the dominant left eigenvector of $\mathbf{P}$, normalised to sum to one. A run whose crisis fraction exceeds 15% is flagged for review.

4.3 Bayesian shrinkage of regime parameters

The raw per-regime moments are stabilised by conjugate shrinkage. With a Gaussian–Gaussian model the posterior mean for regime k is

$$\mu_k^* = w_k \bar{r}_k + (1 - w_k) \mu_0, \quad w_k = \frac{n_k / \sigma^2}{n_k / \sigma^2 + 1 / \tau^2}, \quad (3)$$

where $\mu_0 = \bar{r}$ is the grand mean, σ^2 the pooled within-regime variance, τ^2 the between-regime variance, and n_k the regime's sample size. The weight w_k is the share of total precision coming from data, so sparse regimes shrink harder toward the grand mean. The variance uses an inverse-gamma update,

$$\sigma_k^{*2} = \frac{\beta_{\text{IG}} + \frac{1}{2} SS_k}{\alpha_{\text{IG}} + \frac{n_k}{2} - 1}, \quad SS_k = \sum_{t: s_t = k} (r_t - \bar{r}_k)^2, \quad (4)$$

with prior shape $\alpha_{\text{IG}} = 3$ and $\beta_{\text{IG}} = \sigma^2(\alpha_{\text{IG}} - 1)$, fixing the prior mode at the pooled variance.

4.4 Tail risk via Extreme Value Theory

Let $L_t = -r_t$ be losses. The Hill estimator of the tail index over the top k order statistics is

$$\hat{\xi}_k^{\text{Hill}} = \frac{1}{k} \sum_{i=1}^k [\log L_{(i)} - \log L_{(k+1)}], \quad (5)$$

and the threshold k^* is chosen where $\hat{\xi}_k$ is most stable across a sliding window, searching $k \in [0.02n, 0.15n]$. Given threshold $u^* = L_{(k^*+1)}$ and exceedances $Y_i = L_i - u^*$, the Generalized Pareto parameters $(\hat{\xi}, \hat{\beta})$ are fit by maximum likelihood to $F_u(y) = 1 - (1 + \xi y / \beta)^{-1/\xi}$, yielding

analytical risk measures

$$\widehat{\text{VaR}}_\alpha = u^* + \frac{\hat{\beta}}{\hat{\xi}} \left[(\alpha n / N_u)^{-\hat{\xi}} - 1 \right], \quad \widehat{\text{ES}}_\alpha = \frac{\widehat{\text{VaR}}_\alpha + \hat{\beta} - \hat{\xi} u^*}{1 - \hat{\xi}}, \quad (6)$$

valid for $\hat{\xi} < 1$. If the fit fails or fewer than fifteen exceedances are available the engine falls back to empirical quantiles. Because the input has been winsorised at the 1st/99th percentile, the exceedances are bounded above, which on well-behaved indices drives the fitted shape negative (a short, bounded tail) — a point we return to in Section 6.

4.5 Monte Carlo path generation

Paths are simulated by a Markov regime switch: the next regime is drawn from row k_t of $\mathbf{P}$, and the return is drawn from the shrunken emission law $r_t^{(i)} \mid k_t = k \sim \mathcal{N}(\mu_k^*, \sigma_k^{*2})$. To inject path-dependence, each path's running drawdown state $d_t^{(i)} \in \{0, 1, 2\}$ (normal / moderate / severe, with thresholds -5% and -15%) blends the regime draw with a drawdown-conditional draw estimated from history:

$$r_t^{(i)} \leftarrow (1 - \omega_d) r_t^{(i)} + \omega_d \epsilon_t^{(d)}, \quad \epsilon_t^{(d)} \sim \mathcal{N}(\mu_d, \sigma_d^2), \quad (7)$$

with blend weights $\omega_0 = 0.10$, $\omega_1 = 0.30$, $\omega_2 = 0.50$, so paths in deeper drawdowns draw more heavily on the historical stress distribution. Finally a hard floor enforces simulation integrity: once a path's cumulative return falls below $f = -0.70$, all subsequent returns are set to zero,

$$r_t^{(i)} = 0 \quad \text{if} \quad C_{t-1}^{(i)} \leq f, \quad C_t^{(i)} = \sum_{s \leq t} r_s^{(i)}, \quad (8)$$

reflecting that a position down 70% would in practice be stopped out before compounding further.

4.6 Esscher change of measure

To align the simulated mean and tail with the regime and EVT targets, paths may be reweighted by an exponential (Esscher) tilt,

$$w_i \propto \exp(\lambda_1 \bar{r}_i + \lambda_2 \bar{r}_i \mathbf{1}[\bar{r}_i \leq \widehat{\text{VaR}}_\alpha]), \quad (9)$$

with (λ_1, λ_2) solving $\mathbb{E}_{\mathbb{Q}}[\bar{r}] = \mu^*$ and $\mathbb{E}_{\mathbb{Q}}[\bar{r} \mid \bar{r} \leq \widehat{\text{VaR}}_\alpha] = \text{ES}^*$. The system is solved from five random starts, taking the smallest-residual solution; if that residual exceeds 10^{-4} the tilt reverts to identity weights ($\lambda_1 = \lambda_2 = 0$), preserving the historical measure. Paths are then resampled with replacement in proportion to w_i .

4.7 Stress metrics and bootstrap intervals

From the path set the engine computes the maximum-drawdown distribution, the α -level Expected Shortfall (both aggregate and per-path), and the recovery time distribution. Every scalar θ carries a 90% interval from $B = 500$ path resamples,

$$\widehat{\text{CI}}_{90\%}(\theta) = [\hat{\theta}_{(0.05B)}^*, \hat{\theta}_{(0.95B)}^*]. \quad (10)$$

The bootstrap is fully vectorised so that 500 resamples cost roughly one metric evaluation.

4.8 Model Fragility Index

Definition 1 (Model Fragility Index). Let D_0 be the baseline maximum-drawdown distribution and D_θ the distribution under perturbation θ from a set Θ . Then

$$\text{MFI} = \frac{1}{|\Theta|} \sum_{\theta \in \Theta} \frac{W_1(D_0, D_\theta)}{\sigma(D_0)}, \quad W_1(\mu, \nu) = \int_0^1 |F_\mu^{-1}(u) - F_\nu^{-1}(u)| du, \quad (11)$$

where W_1 is the Wasserstein-1 distance and $\sigma(D_0)$ normalises by the baseline drawdown spread, making the index scale-invariant.

The perturbation set comprises five independent shocks — regime means $\times 1.10$ and $\times 0.90$, regime volatilities $\times 1.20$ and $\times 0.80$, and a 5% Dirichlet jitter of the transition rows — each run under the baseline seed so that only the parameter change, not sampling noise, moves the output. Grades follow $\text{MFI} < 0.25$ Robust, $0.25 \leq \text{MFI} < 0.65$ Moderate, $\text{MFI} \geq 0.65$ Fragile.

4.9 Multi-asset extension

For portfolios the engine estimates a Ledoit–Wolf shrinkage covariance, draws correlated regime innovations through a Cholesky factor, and aggregates path P&L at supplied weights, so joint drawdowns reflect both marginal tails and cross-asset dependence. The validation in this paper is single-asset; the multi-asset path is left to future work.

5. System Architecture and Implementation

The engine ships as a production service, not only a research script. The numerical core (`engine/core.py`) is pure NumPy/SciPy and has no web dependencies. A FastAPI application (`api/`) wraps it with token- and API-key authentication, submits long runs as background jobs, and persists runs and results through an async SQLAlchemy layer (`db/`). Results are converted to a JSON-safe payload by a dedicated serialiser (`engine/serializer.py`) and to a formatted tear-sheet by a ReportLab module (`reports/pdf.py`); both are reachable from one-click buttons in the user interface. Two front-ends consume the API: a React single-page

application (`web/`) for interactive use and a Streamlit app (`frontend/`) for rapid analysis. The same code runs locally under Docker Compose and in the cloud, and every run is reproducible from its seed.

6. Single-Asset Characterisation: SPY

We first characterise the engine on the SPDR S&P 500 ETF (SPY), daily from January 2010 to June 2026, 4,138 observations, 17.1% annualised volatility, 10,000 paths over a 252-day horizon.

Regimes. The detector places 70.1% of days in calm, 24.0% in volatile, and 5.8% in crisis; the long-run stationary distribution is (0.70, 0.24, 0.06). A crisis share near 6% is consistent with the observed frequency of sustained equity stress over the period (chiefly the March 2020 crash and the 2022 tightening cycle).

Tail. The Peaks-over-Threshold step used 275 exceedances beyond a daily loss threshold of 1.43%, with an implied 5% Expected-Shortfall target of -2.79% . As anticipated in Section 4, winsorising before the fit bounds the exceedances and yields a negative GPD shape ($\hat{\xi} = -1.26$, $\hat{\beta} = 0.020$): a short, bounded tail relative to the threshold rather than a Fréchet-type fat tail. For this sample the Esscher solver did not improve on the historical measure and reverted to identity weights, so the metrics below are reported under $\mathbb{P}$.

Risk metrics. Table 1 reports the core outputs with 90% bootstrap intervals. The mean one-year maximum drawdown is -14.1% and the 5th-percentile drawdown is -37.0% , a sensible tail benchmark for position sizing on a 17%-vol index. The high never-recover rate is a horizon artefact: many paths that draw down late in the 252-day window simply run out of time to recover, not a statement about the long-run index.

Table 1: SPY risk metrics: 10,000 paths, 252-day horizon, seed 42.

Metric	Estimate	90% CI lower	90% CI upper
Mean max drawdown	-14.1%	-14.2%	-13.9%
Median max drawdown	-11.0%	—	—
5th-percentile drawdown	-37.0%	-37.8%	-36.3%
Aggregate ES (5%)	-1.80%	-1.81%	-1.80%
Median recovery time	35 days	—	—
Never recover (252d)	53.2%	—	—
Model Fragility Index	0.25	Moderate	

Fragility. The MFI of 0.25 sits on the Robust/Moderate boundary. Volatility perturbations dominate the index (W_1 of 0.043 and 0.041 for $\pm$ shocks, versus 0.005–0.008 for mean shocks

and 0.033 for transition jitter), as expected since the drawdown distribution is driven primarily by the volatility of the return process.

7. Out-of-Sample Backtest and Verification

The single-asset section shows the engine produces internally sensible numbers. The harder question is whether its predicted distributions actually contained what happened. We answer it out-of-sample.

7.1 Protocol

For an asset with returns $r_{1:T}$ and dates $d_{1:T}$ and a crisis window $[t_0, t_1]$, the *training set* is $\{r_i : d_i < t_0\}$ — strictly pre-crisis — and we require at least 504 training observations. The engine is fit on the training set (sensitivity and its internal validator disabled, seed 42, $N = 5000$ paths) with horizon H equal to the number of trading days in $[t_0, t_1]$, so the predicted maximum-drawdown distribution and the realized maximum drawdown span the same length. All draw-downs are computed in additive cumulative-return space, the engine’s internal convention, applied identically to simulated and realized paths.¹

7.2 Coverage statistics

For each cell we record the realized maximum drawdown DD^{real} and the simulated drawdown distribution $\{DD^{(j)}\}_{j=1}^N$, and compute the *probability-integral-transform rank*

$$u = \frac{1}{N} \sum_{j=1}^N \mathbf{1}[DD^{(j)} \leq DD^{\text{real}}], \quad (12)$$

the fraction of simulated paths at least as bad as reality. Under a correct model $u \sim \text{Uniform}(0, 1)$. A $p5$ ($p1$) breach occurs when DD^{real} is worse than the simulated 5th (1st) percentile. For x breaches in n cells at level p the Kupiec proportion-of-failures statistic

$$\text{LR}_{\text{POF}} = -2 \ln \frac{(1-p)^{n-x} p^x}{(1-\hat{\pi})^{n-x} \hat{\pi}^x}, \quad \hat{\pi} = x/n, \quad (13)$$

is asymptotically χ_1^2 . The calm control provides the regime in which the test’s nominal-rate assumption is appropriate; on the deliberately adversarial crisis set the statistic quantifies the conditional miss rather than testing for “pass.”

¹The additive maximum drawdown of a path is $\min_t (C_t - \max_{s \leq t} C_s)$ with $C_t = \sum_{i \leq t} r_i$. Because it does not compound it can fall below -100% in extreme cases; in price terms the figures are smaller (e.g. financials in 2008 troughed near -83%).

7.3 Data universe

Ten liquid proxies span equity beta (SPY, QQQ, IWM, DIA, EFA), sector concentration (XLF, XLE), duration (TLT), credit (HYG), and a diversifier (GLD), each pulled to the report date through the engine’s own loader. Nine crisis windows run from the 2008 financial crisis through the 2023 regional-bank episode. The grid yields 89 crisis cells (one skipped: high-yield credit in 2008, which lacks the required pre-crisis history) and 213 calm-control cells from 22 non-overlapping six-month windows.

7.4 Calm-period calibration

Table 2 and Figure 1 establish the control. On calm windows the engine breaches its 5% bound in 2.8% of cells and its 1% bound in 0.9% — both statistically indistinguishable from nominal (Kupiec $p = 0.11$ and 0.93). The mean PIT rank of 0.69 shows the engine is, if anything, mildly conservative on ordinary data, predicting somewhat more risk than realises. This is the prudent direction for a risk system and the precondition for taking the crisis results seriously.

Table 2: Headline coverage: calm control versus crisis event study.

	Calm control	Crisis study	<i>Calibrated target</i>
Cells evaluated	213	89	—
$p5$ breach rate	2.8%	56.2%	$\approx 5\%$
Kupiec p -value	0.11	$< 10^{-40}$	> 0.05
$p1$ breach rate	0.9%	31.5%	$\approx 1\%$
Kupiec p -value	0.93	$< 10^{-33}$	> 0.05
90% band coverage	71.8%	42.7%	$\approx 90\%$
Mean PIT rank	0.69	0.14	0.50

7.5 Crisis event study

The crisis picture (Table 3, Figures 2–3) inverts the control. Coverage degrades monotonically with severity: the 2015–16 China/oil selloff is fully covered and the 2023 regional-bank episode largely so, while every COVID-2020 cell and nearly every 2008 cell breach even the 1% bound. Realized crisis drawdowns sit at a mean PIT rank of 0.14 — deep in the predicted lower tail — and the two great systemic shocks land essentially at rank zero.

7.6 Cross-asset behaviour

Sorting by mean PIT rank (Table 4) shows the gap concentrates in high-beta and cyclical exposures — energy, small caps, developed-international, broad equity — which suffer the most violent crisis drawdowns. Defensive and diversifying sleeves behave very differently: long-duration Treasuries (rank 0.30) and gold (rank 0.35) sit near the centre of the predicted distribution, and

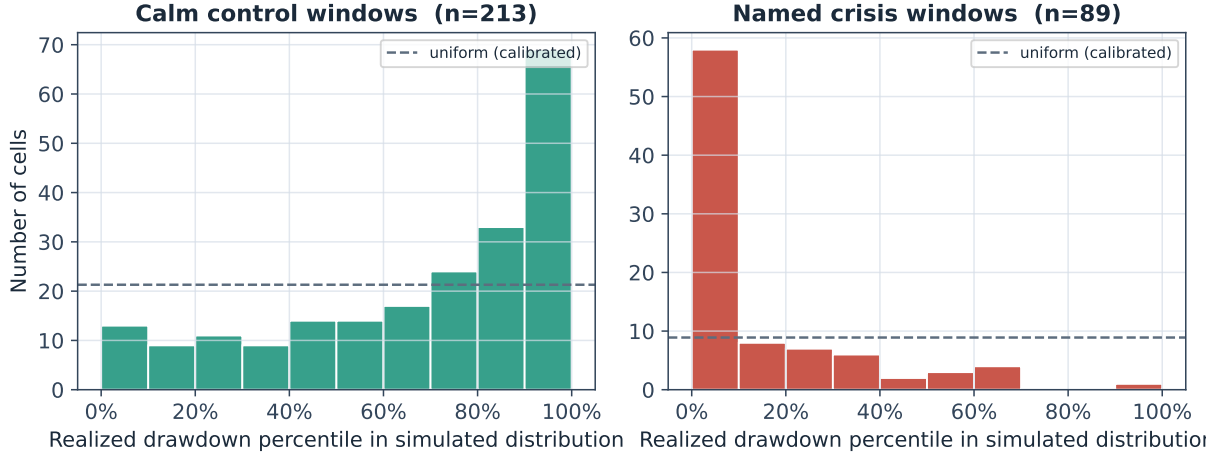


Figure 1: Realized-drawdown PIT ranks. Calm windows (left) track the uniform benchmark; crisis windows (right) pile at the left edge, where realized losses fall in the engine’s lower tail.

Table 3: Per-crisis coverage. “Mean DD” and “Pred. $p5$ ” are cross-asset averages (additive); “Rank” is the mean realized PIT rank (lower = deeper miss).

Crisis	n	Mean DD	Pred. $p5$	$p5$ br.	$p1$ br.	Cov.	Rank
2010 Flash Crash / Euro I	10	−14.3	−17.2	4/10	0/10	60%	0.20
2011 US Downgrade / Euro II	10	−19.4	−16.7	7/10	2/10	30%	0.09
2015–16 China / Oil Selloff	10	−16.8	−24.6	0/10	0/10	100%	0.27
Feb 2018 Volmageddon	10	−8.5	−6.7	8/10	3/10	20%	0.06
Q4 2018 Selloff	10	−18.5	−17.0	6/10	4/10	30%	0.19
March 2020 COVID Crash	10	−37.7	−9.9	10/10	10/10	0%	0.00
2022 Rate-Shock Bear	10	−27.7	−29.6	4/10	1/10	60%	0.09
2023 Regional Bank Crisis	10	−4.8	−7.4	3/10	0/10	70%	0.32
2008 Global Financial Crisis	9	−52.2	−26.4	8/9	8/9	11%	0.01

high-yield credit is breached in only two of eight cells. The miss is a faithful reflection of which exposures carry systemic tail risk, not uniform model error.

7.7 Exemplar cells

Four cells make the spectrum concrete. **SPY in COVID-2020**: realized −38.2% against a predicted $p5$ of −8.6% ($p1$ −11.9%), rank 0.000 — the fastest 30% equity drawdown in history lies outside anything pre-2020 data implied. **XLFF in 2008**: realized −103.6% (additive) against $p5$ −38.2%, rank 0.000 — a solvency crisis in the financial sector itself. **SPY in 2015–16**: realized −13.3% against $p5$ −20.3%, rank 0.25 — a textbook covered correction. **SPY in 2023**: realized −3.4% against $p5$ −6.7%, rank 0.28 — contained, and the engine knew it.

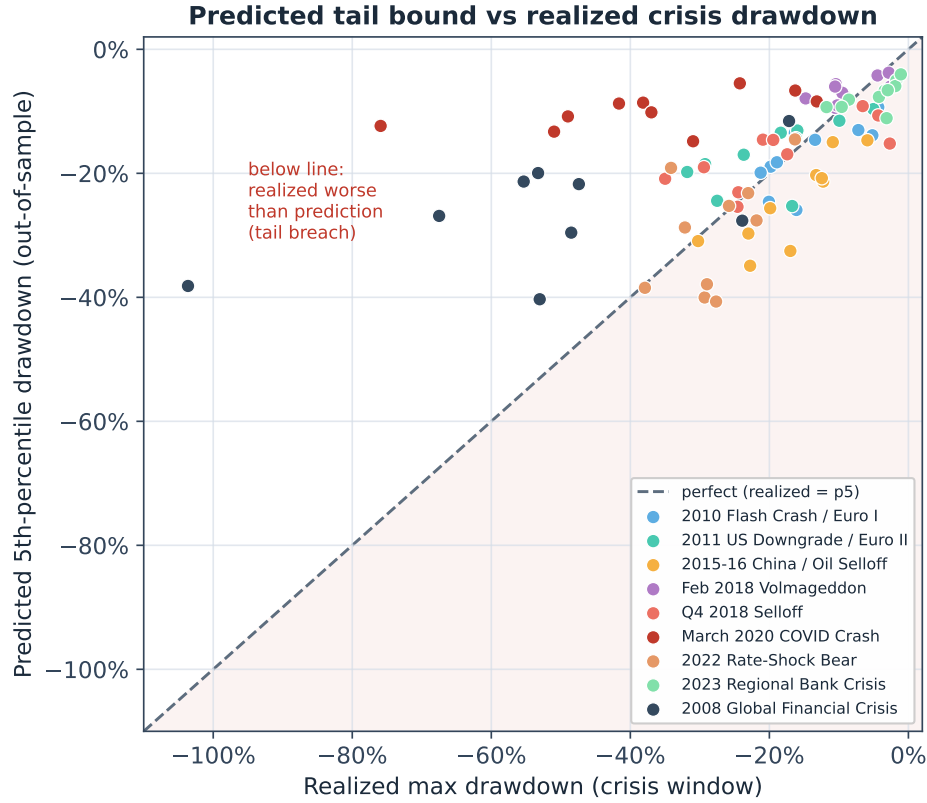


Figure 2: Predicted 5th-percentile drawdown versus realized crisis drawdown, one point per cell. Points below the identity line are tail breaches; the lower-left cluster is COVID-2020 and 2008.

8. Discussion

The two regimes deliver one message. The calm control shows the engine is not broken: at both the 5% and 1% tail levels its out-of-sample exception rates are statistically indistinguishable from nominal, and it errs toward caution. The crisis study shows that the outcomes which *do* breach are exactly the events a history-only model must miss — the sharpest, most reflexive systemic shocks, where volatility, correlation, and liquidity dislocate faster than any trailing window can encode. Before March 2020 there was no 2020 in the data.

This is a property of the estimation problem, not a defect unique to this engine. Any model whose tail is calibrated to realized returns inherits the ceiling of its worst observed history. The contribution of the backtest is to *measure* the residual gap rather than hide it, and to show that the gap is monotone in severity and concentrated in the exposures that genuinely carry systemic risk. Operationally this is why the production engine does not rely on the empirical tail alone: its crisis-regime structure, drawdown-conditional blend, and the operator-tunable loss floor exist to extend the simulated distribution into the region this backtest shows lies beyond historical reach. The honest deployment is to use the engine as calibrated here for ordinary risk, with the

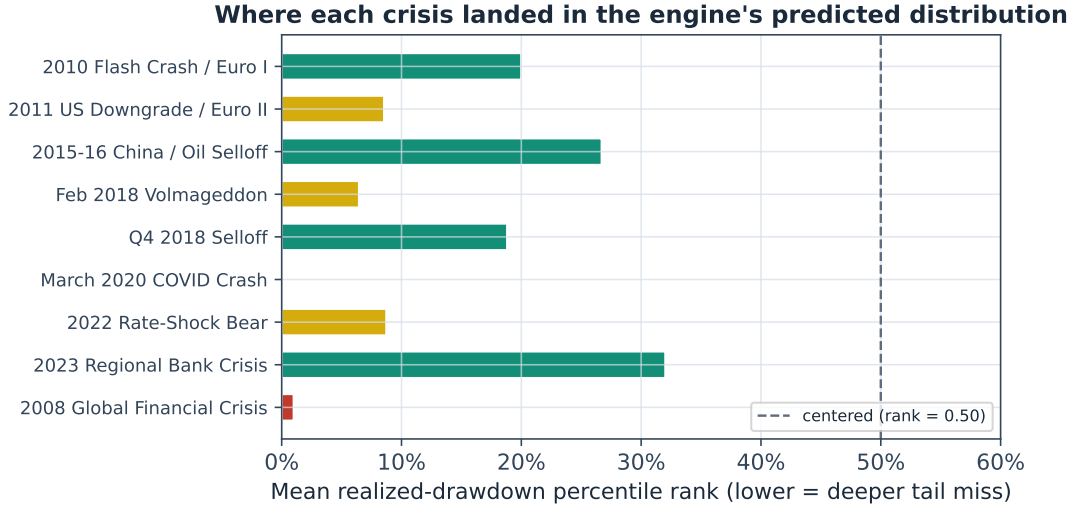


Figure 3: Mean realized-drawdown percentile rank by crisis. Bars near 0.50 mean the engine centred the outcome; bars near zero mean it fell in the extreme tail.

Table 4: Per-asset coverage across crises, deepest to shallowest tail miss.

Asset	n	Mean DD	Pred. $p5$	$p5$ br.	Cov.	Rank
Energy (XLE)	9	-32.9	-20.9	7/9	22%	0.03
Russell 2000 (IWM)	9	-29.4	-20.0	7/9	22%	0.04
Developed ex-US (EFA)	9	-24.2	-17.0	7/9	22%	0.06
S&P 500 (SPY)	9	-22.2	-14.2	7/9	22%	0.07
Financials (XLF)	9	-33.2	-24.2	6/9	33%	0.07
Dow 30 (DIA)	9	-20.7	-15.0	5/9	44%	0.10
High-Yield Credit (HYG)	8	-9.9	-9.6	2/8	75%	0.16
Nasdaq 100 (QQQ)	9	-23.3	-24.8	3/9	67%	0.19
20Y+ Treasuries (TLT)	9	-10.4	-10.1	4/9	56%	0.30
Gold (GLD)	9	-11.2	-15.0	2/9	67%	0.35

crisis overlay engaged when sizing for systemic scenarios.

9. Limitations

1. **Adversarial crisis sample.** The nine windows are chosen because they are extreme, so the crisis breach rates estimate a conditional, not an unconditional, exception rate; the unconditional estimate is the calm control.
2. **Additive drawdown convention.** Drawdowns are summed, not compounded, to match the generator; this inflates very deep drawdowns relative to price terms but is applied identically on both sides of every comparison.
3. **Horizon matching.** Setting H to the realized window length is the cleanest like-for-like

choice but evaluates short, sharp episodes over short horizons where the simulated distribution is itself narrow.

4. **Numerical fallbacks.** The Esscher tilt and the GPD fit both fall back to conservative defaults (identity weights, empirical quantiles) when their solvers do not converge; these events are logged.
5. **Single data vendor.** All prices come from one adjusted-close source; vendor differences in corporate-action handling would shift individual cells modestly but not the aggregate picture.
6. **Univariate validation.** The backtest is single-asset; the multi-asset dependence machinery is described but not validated here.

10. Conclusion and Future Work

The Blue Lotus engine turns a return series into a fully characterised, bootstrap- bounded distribution of drawdown, tail-loss, and recovery outcomes, built from auditable components and shipped as a reproducible production service. Under a strict out-of-sample protocol it is statistically well calibrated on ordinary market conditions — unrejectable at either the 5% or 1% tail level across 213 calm cells — and its behaviour under the nine largest crises of the past two decades is fully characterised, with coverage that degrades monotonically in severity and a clean break only at the two great systemic shocks.

Three extensions follow naturally. The regime detector could ingest exogenous stress signals — implied volatility, credit spreads, funding indicators — to sharpen crisis identification ahead of realised volatility. The multi-asset dependence model deserves the same out-of-sample treatment given here to the univariate case, validated against joint drawdown events. And the Model Fragility Index could itself be backtested, by asking whether high-fragility periods precede realised instability. In every case the standard set here is the same: measure the gap, and report it.

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A. Default Parameters

Table 5: Engine default parameters used throughout this paper.

Module	Parameter	Default
Preprocessing	winsorisation	1st / 99th percentile
Regime	rolling window w	20 days
Regime	vol thresholds	70th / 90th percentile
Regime	min crisis duration	5 days
Shrinkage	IG prior shape	$\alpha_{IG} = 3$
Tail	EVT level α	0.05
Tail	Hill search range	$[0.02n, 0.15n]$
Monte Carlo	paths N	10,000 (5,000 in backtest)
Monte Carlo	horizon	252 days (window length in backtest)
Monte Carlo	drawdown blend	0.10/0.30/0.50
Monte Carlo	loss floor f	−0.70
Esscher	residual tolerance	10^{-4}
Bootstrap	resamples B	500
MFI	grades	0.25 / 0.65
All	random seed	42

B. Reproducibility

The single-asset characterisation, the crisis event study, the calm control, the aggregate tables, and the figures all regenerate from `backtest/oos_backtest.py`, `backtest/control_backtest.py`,

`backtest/aggregate.py`, and `backtest/make_figures.py` with fixed seed 42. Raw per-cell output is written to `backtest/results/` as JSON.